

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-5955
Project Name	PocketSmart AI - Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Overview

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4 spaces used, PEP8 followed
2	Proper File Structure	Files and folders are logically organized	Yes	routes/, services/, models/, templates/, static/
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	user_budget, home_preferences, jewelry_style, outfit_image
4	Function/Method Names	Functions are descriptively named	Yes	generate_home_recommendations(), get_jewelry_match()
5	Code Comments	Inline and block comments explain logic	Yes	Docstrings for each route, comments for Gemini prompts
6	Modular Design	Code is split into reusable functions/modules	Yes	gemini_utils.py reusable for all 3 planners
7	No Redundant Code	Duplicate or unused code is removed	Yes	Removed mock data, kept DRY principle
8	Error Handling	Exceptions and errors are handled gracefully	Yes	try/except for API failures, fallback recommendations

Page 2 — Code Layout, Readability and Reusability for PocketSmart AI

Reusable Components / Modules:

S.No	Component/Module Name	Language/Technology	Where Reused	Reusability Level (High/Medium/Low)
1	gemini_utuls.py	Python / google-generativeai	Home, Party, Jewelry planners - generate recommendations	High
2	Auth Module /auth	Python Flask, JWT, bccrpt	Login, Register, /token, /session-info, /session-data	High
3	Budget Allocation Logic	Python	/generate-home, /generate-party proportional split	High
4	Image Processing Module	Python Pillow	Jewelry planner outfit image analysis for matching	Medium
5	Template Rendering /history	Flask Jinja2 HTML/CSS/JS	Dashboard, Home/Party/Jewelry results, history page	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments	
Code Layout & Structure	5	Well-structured modular Flask/FastAPI, routes/, services/, models/ separation	5/5 ★★★★★★
Readability	5	Meaningful names, PEP8, docstrings, comments, clean code	5/5 ★★★★★★
Reusability	5	gemini_utuls reusable, auth reusable, budget logic reused across planners	5/5 ★★★★★★
Documentation/Comments	4	API docs, inline comments, README, prompt explanations	4/5 ★★★★☆
Overall Score	5	Maintainable, scalable, extensible for future categories like travel planning	5/5 ★★★★★